Age-Optimal Target Wake Time: Provably Good Wake Schedules for Energy-Constrained Wi-Fi Status Updating

Haoyu Wang*, Bo Sheng*, Xiaoqian Zhang†
*University of Massachusetts Boston, Boston, MA, USA
{haoyu.wang001, bo.sheng}@umb.edu
†University of Nebraska Omaha, Omaha, NE, USA
xiaoqianzhang@unomaha.edu

***Abstract*—Target Wake Time (TWT), introduced in IEEE 802.11ax, lets an access point schedule exactly when each station wakes, transmits, and dozes. Existing TWT schedulers optimize energy or throughput, treating information freshness at best as a constraint and offering no performance guarantees. We design the wake schedule itself for freshness: minimize the weighted average Age of Information (AoI) over stations subject to per-station energy budgets, where the decision variables are the TWT triples (wake interval, offset, service period duration). We derive a renewal-exact AoI model for TWT under per-SP block fading and validate it against packet-level 802.11ax simulation with $\sim$1% mean error. We show that, unlike preemptive scheduling, non-preemptive TWT packing can be infeasible at schedule density 1, and identify the granularity condition under which a small-first best-fit packer provably succeeds. Around this we build HARMONIC-GREEDY, a scheduler combining a convex relaxation, anchor-optimized power-of-two rounding, and a best-of-uniform safeguard, and prove it is a constant-factor approximation: $4/\ln 2 \approx 5.77$ under a mild granularity assumption and $6/\ln 2 \approx 8.66$ unconditionally. We implement the complete system in ns-3 – a TWT wake/doze mechanism integrated with the power-save architecture, plus the scheduler – and show that it is the only scheduler that stays near a relaxation lower bound across all regimes: against a strong energy-greedy baseline it ties when per-station energy floors already pin the periods, and wins by 5–36% exactly where the schedule density is binding and must be redistributed by AoI weight or channel quality rather than by energy budget – the regime our analysis identifies.**

***Index Terms*—Age of Information, Target Wake Time, IEEE 802.11ax, scheduling, energy efficiency, ns-3**

I. INTRODUCTION

Real-time monitoring applications – industrial sensing, health wearables, smart-building telemetry – send small periodic status updates over Wi-Fi while running on batteries. Two trends collide in such systems. On one hand, the relevant performance metric is not throughput or even delay, but the *Age of Information* (AoI) [1], [2]: the time elapsed since the generation of the freshest delivered update. On the other, the dominant energy cost is simply being awake: a station that wakes rarely is efficient but stale; one that wakes often is fresh but dies quickly.

IEEE 802.11ax [3] introduced Target Wake Time (TWT) precisely to manage this trade-off mechanically: the AP and a station agree on a periodic schedule of *service periods* (SPs) – a wake interval T, an offset ϕ, and an SP duration d – and the station dozes outside its SPs. TWT thus exposes, for the first time in mainstream Wi-Fi, the exact control surface that AoI theory wants to optimize: *when* each source is allowed to deliver. Yet existing TWT schedulers optimize energy or throughput, treat freshness at best as a constraint, and offer no guarantees (Section II).

This paper designs the TWT schedule itself for freshness, with guarantees. Our contributions:

- **Model** (Sections III–IV): a renewal-exact expression for time-average and peak AoI of a TWT station under per-SP block fading, including the per-wake overhead c_w. A structural lemma shows batching retransmission attempts within an SP is *never* beneficial without wake overhead – c_w is what makes TWT scheduling a genuinely new problem rather than abstract sampling.
- **Hardness structure** (Section V): a three-station example with schedule density exactly 1 that is infeasible under *every* placement – non-preemptive TWT packing fundamentally differs from preemptive utilization bounds (harmonic rate-monotonic is optimal at density 1 only because preemption can split service across gap copies).
- **Algorithm and guarantees** (Sections VI–VII): HARMONIC-GREEDY, combining a water-filling convex relaxation, power-of-two period rounding with an anchor chosen by a candidate-argmin rule, small-first best-fit packing of dyadic reservations, and a best-of-uniform safeguard. We prove a deterministic $4/\ln 2 \approx 5.77$ approximation under a mild granularity assumption, and $6/\ln 2 \approx 8.66$ unconditionally. The packing analysis gives a tight granularity threshold $d \leq t_0(1-U)$.
- **System** (Section VIII): a complete ns-3 implementation – a TWT wake/doze mechanism integrated with the 802.11 power-save architecture, with SP-boundary queue gating and TSF-anchored offsets – validated against the analytical model with $\sim$1% mean error, plus five protocol findings about deploying TWT (e.g., doze announcements cannot be acknowledged over a faded channel; pending

retransmissions must be gated or they erase the energy savings).

- **Evaluation** (Section IX): against both a naive (equal-interval) and a strong (energy-greedy) baseline, plus a relaxation lower bound. We separate two effects – period diversity, which the energy-greedy baseline already captures, and AoI-weighting, which we isolate in uniform-budget binding-density regimes where only our scheduler wins (5–6.5% over both baselines, up to 36% at scale with mixed modulation rates), exactly where the theory predicts.

II. Related Work

Age of Information. Since its introduction as a freshness metric [1], AoI has grown into a broad field spanning queueing, sampling, and estimation [2], [4]. Foundational results characterize the freshness-optimal update policy for a single source [5], the effect of packet management and queue discipline [6]–[8], and AoI's role in IoT [9]. Variants such as the Age of Incorrect Information [10] refine the metric for estimation. Our renewal model (Sec. IV) specializes this machinery to the TWT service-period structure.

AoI scheduling. Minimizing weighted AoI across many sources over a shared channel is the closest body of work. Centralized per-slot policies and their optimality were established for broadcast networks with reliable and unreliable links [11], [12] and with stochastic arrivals [13], [14]; the restless-bandit structure yields Whittle-index policies that are near-optimal and, in cases, asymptotically optimal [15]. These works schedule *which source transmits in each slot* under a centralized controller; we instead design *periodic wake agreements* fixed in advance, where a sleeping station is unreachable between its service periods – a constraint absent from slot-by-slot models. A parallel line constrains energy explicitly – energy-harvesting sources [16], [17], retransmission/ARQ budgets [18], and distributed freshness optimization [19] – but optimizes transmission timing rather than a negotiated periodic wake schedule with a hard duty-cycle budget. Closest in spirit are sleep-wake AoI optimization for CSMA sensors [20], which tunes continuous wake *rates* under battery constraints but has no negotiated periodic structure, and age-agnostic cyclic schedulers [21], which optimize deterministic transmission patterns but model neither energy nor sleeping.

TWT and Wi-Fi. TWT was introduced in IEEE 802.11ax to manage energy and contention [3], [22], [23], and is extended in Wi-Fi 7 (802.11be) – notably restricted TWT for latency-sensitive traffic and multi-link operation [24], [25] – making schedule design increasingly central [26]. TASPER [27] schedules TWT SPs to maximize throughput and energy efficiency subject to AoI *constraints*, proves NP-hardness, and gives a heuristic without guarantees; experimental IIoT schedulers [28] respect AoI deadlines deterministically; and TWT mechanism implementations and measurement studies [29]–[31] provide the platform context. To our knowledge no prior work makes AoI the *objective* of TWT schedule design or provides approximation guarantees.

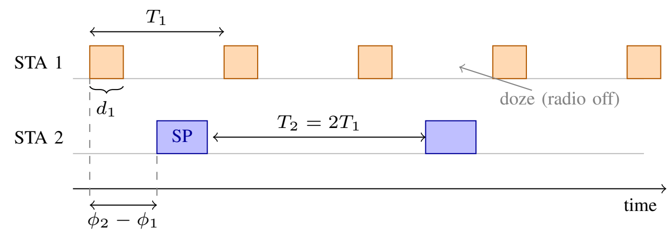

Fig. 1. TWT wake schedules: station i wakes for d_i every T_i at offset ϕ_i on a common (TSF-anchored) grid; the scheduler chooses the triples so that service periods are disjoint.

Periodic real-time scheduling. Our packing analysis connects to classical real-time theory. Rate-monotonic scheduling is optimal for preemptive periodic tasks [32], and harmonic task sets are schedulable up to utilization 1 – but only because preemption can split a job across the gaps left by higher-rate tasks. TWT service periods are *contiguous and non-preemptive*, which is exactly why our Proposition 2 shows density 1 can be infeasible. The relevant non-preemptive analogue is the pinwheel problem and distance-constrained scheduling [33]–[35], whose density thresholds and harmonic constructions inform our small-first best-fit packer; we adapt them to the AoI-weighted, energy-budgeted TWT setting.

III. System Model

N stations are associated with one AP. Station i holds a TWT agreement (T_i, ϕ_i, d_i): it wakes at times $\phi_i + kT_i$, stays awake for d_i, and dozes otherwise (Fig. 1). SPs of different stations must not overlap (the scheduler's responsibility; overlaps cause contention we wish to design away). Station i samples its source at the start of each SP (generate-at-will) and transmits the update uplink within the SP.

Channel. We model per-SP block fading: the channel coherence time exceeds the SP duration, so all transmission attempts within one SP share fate; an SP succeeds with probability $p_i \in (0, 1]$, independently across SPs. This is the regime in which retransmission policy matters for AoI: with per-attempt i.i.d. loss, in-SP retries drive the per-SP success probability to 1 and the problem degenerates (we verify this empirically in Section IX).

Energy. The station's energy cost is its awake time: each wake costs a fixed overhead $c_{w,i}$ (radio ramp-up, synchronization) plus the SP airtime. Station i has a duty-cycle budget ρ_i: $d_i/T_i \le \rho_i$, equivalently $T_i \ge T_i^{\min} := d_i/\rho_i$.

Objective. Let $A_i(t)$ be the age of the freshest update from station i delivered to the AP. We minimize the weighted time-average AoI $\sum_i w_i \bar{A}_i$; Section VII notes that all results carry to weighted peak AoI.

Scope. We deliberately study the cleanest setting that isolates schedule design: uplink status updates, generate-at-will sampling, scheduled non-overlapping SPs (so contention is designed away), and per-SP block fading. These assumptions are standard in AoI scheduling [11], [21] and matched to monitoring IoT, the dominant TWT use case. Random arrivals couple the per-station decisions through the density constraint and turn the design into a constrained-MDP problem; downlink AoI requires broadcast TWT and AP-side delivery; both are natural extensions we discuss in Section X but do not treat here.

IV. Single-Station Structure

Successful SPs form a renewal process; the standard renewal-reward AoI argument [1], [2] then gives the following, which is the linchpin that turns TWT scheduling into a linear objective (Section V).

Proposition 1 (TWT AoI)**.** *With wake interval T, per-SP success probability p, and mean in-SP delivery delay δ,*

$$\bar{A} = \delta + T\Big(\frac{1}{p} - \frac{1}{2}\Big), \qquad \bar{A}^{\text{peak}} = \delta + \frac{T}{p}.$$

Sketch. Let G be the number of wake intervals between successive successful SPs. Since each SP succeeds independently with probability p, G is geometric on $\{1, 2, \dots\}$, so $\mathbb{E}[G] = 1/p$ and $\mathbb{E}[G^2] = (2-p)/p^2$. The inter-delivery time is $X = GT$. After a successful delivery the AoI drops to the in-SP delay (mean δ) and then grows linearly until the next success, so each renewal cycle contributes area $\delta X + X^2/2$. Dividing expected area by expected cycle length gives $\bar{A} = \delta + \mathbb{E}[X^2]/(2\mathbb{E}[X]) = \delta + T(1/p - 1/2)$, and the per-cycle peak $\delta + X$ averages to $\bar{A}^{\text{peak}} = \delta + \mathbb{E}[X] = \delta + T/p$. □

When the channel is per-attempt i.i.d. with success probability q and the SP holds k attempts, $p = 1 - (1-q)^k$ and $d = ks + c_w$ with s the per-attempt airtime. Choosing k under the energy-tight period $T = d/\rho$ reveals a clean dichotomy:

Lemma 1 (batching needs overhead)**.** *With $c_w = 0$, the single attempt $k^* = 1$ is optimal for every q: writing $f(k) = (k + c_w/s)(1/p(k) - 1/2)$, one computes $f(2) - f(1) = q/(2(2-q)) > 0$. With $c_w > 0$, $k^* > 1$ for sufficiently large c_w/s at any fixed q.*

The per-wake overhead is thus load-bearing: it is exactly what distinguishes TWT schedule design from abstract sampling models, and we keep it in the model throughout.

We validated Proposition 1 against packet-level 802.11ax simulation (Section VIII): measured time-average AoI matches within 0.02% ($p \approx 1$, two parameter sets) and within 0.6–1.2% under per-SP block fading with $p \in \{0.8, 0.6, 0.4\}$, with measured duty cycles equal to d/T to four decimals.

V. The Multi-Station Problem

By Proposition 1 the schedule cost is $J(T) = \sum_i a_i T_i + C$ with $a_i := w_i(1/p_i - 1/2)$ and C a T-independent constant; the design problem is

$$\min_{T,\phi} \ \textstyle\sum_i a_i T_i \qquad \text{s.t. SPs pairwise disjoint,} \quad T_i \geq T_i^{\min}. \tag{1}$$

Disjointness implies the *density* bound $U = \sum_i d_i/T_i \leq 1$, but the converse fails in a structural way:

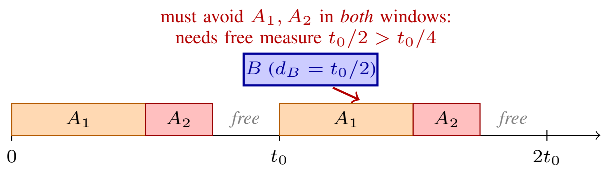

Fig. 2. Proposition 2: at density 1, the per-window free set has measure $t_0/4$, but B's contiguous SP needs $t_0/2$ of it – in every window, under every placement.

Algorithm 1 Harmonic-Greedy

1. per station, choose attempts k_i (Lemma 1) and $d_i \leftarrow k_i s_i + c_{w,i}$
2. solve relaxation (2) with density target $\rho^\star$ (water-filling, Lemma 3)
3. round periods up to grid $\{t_0 2^j\}$, choosing t_0 by candidate-argmin (Corollary 1)
4. re-expand SPs into rounding slack when $p_i < 1$ (more attempts at fixed T are free for AoI)
5. pack by small-first best-fit (Section VII); emit (T_i, ϕ_i, d_i)
6. **safeguard**: if the uniform schedule $T_u = \max(\max_i T_i^{\min}, \sum_i d_i)$ has lower estimated cost, return it

Proposition 2 (density 1 can be unpackable)**.** *Let t_0 be a base period and consider stations A_1: (T=t_0, d=$t_0/2$), A_2: (T=t_0, d=$t_0/4$), B: (T=$2t_0$, d=$t_0/2$), with $U = 1$. No placement of offsets makes the SPs disjoint.*

Proof. A_1, A_2 repeat in every base window, leaving a free set S of measure $t_0/4$, identical in every window. B's SP is a contiguous interval of length $t_0/2$ meeting at most two consecutive windows, as a suffix of one and a prefix of the next; both pieces must lie in S, so $|S| \geq t_0/2$, a contradiction (Fig. 2). □

Remark 1. Under *preemptive* scheduling the same instance is feasible (harmonic rate-monotonic is optimal at utilization 1): preemption splits B's service across the two gap copies. TWT SPs are contiguous by design, so utilization intuition from real-time scheduling does not transfer. Any packing guarantee must involve granularity, which our analysis makes precise (Lemma 5).

A second structural fact bounds every period from below:

Lemma 2 (long SPs dominate periods)**.** *In any feasible schedule, $T_i \geq d_{\max} := \max_k d_k$ for all i.*

Proof. If $T_A < d_B$, the starts of A's SPs are spaced T_A apart, so one lies inside B's SP, and the corresponding A-SP overlaps it. □

VI. The Harmonic-Greedy Scheduler

The relaxation replaces disjointness by its density consequence:

$$J^* = \min\Big\{ \sum_i a_i T_i : \sum_i \frac{d_i}{T_i} \le 1,\ T_i \ge \max(T_i^{\min}, d_{\max}) \Big\}, \tag{2}$$

valid as a lower bound on (1) by Lemma 2.

Lemma 3 (water-filling)**.** *(2) is convex; its solution is $T_i(\lambda) = \max\big(T_i^{\min}, d_{\max}, \sqrt{\lambda d_i / a_i}\big)$ with $\lambda \ge 0$ the density multiplier (found by bisection).*

Two implementation-driven design elements deserve emphasis, both discovered by simulation. First, the *anchor matters*: rounding to powers of two of an arbitrary base can double a period; choosing the anchor among the ideal periods folded into $(T^{\min}/2, T^{\min}]$ – each candidate makes one station's rounding lossless – is provably as good as a random anchor (Corollary 1) and empirically much better. Second, the *safeguard matters*: when the relaxation's period spread is sub-octave (all ideal periods within a factor 2), the harmonic grid cannot express the differentiation and rounding losses dominate. For example, ideal periods of 30 and 50 ms (ratio $1.67 < 2$) round to the *same* power-of-two grid level, so the harmonic schedule cannot give the more important station a shorter period and merely pays the rounding cost; the safeguard then returns the uniform schedule instead. We call this the *octave quantization* effect and quantify it in Section IX.

VII. Approximation Guarantees

We analyze a conservative variant: density target $\rho^\star$, dyadic *reservations* (each station reserves the smallest $t_0/2^m \ge d_i$; it transmits only d_i inside the reservation, so reservations cost no energy), and small-first best-fit (SF-BF) packing: process stations by increasing period; place each into a minimum-size free block of sufficient size, carving from the left and returning the staircase remainder.

Lemma 4 (count invariant)**.** *During SF-BF, new free blocks are only created at sizes currently absent (the staircase sizes lie in $[q, b)$, empty by minimality of the chosen block b); at the start of level-j processing every size has at most 2^j free blocks. Both bounds are tight.*

Lemma 5 (packing threshold)**.** *If every reservation satisfies $d_i' \le t_0(1-U)$ with $U = \sum_i d_i'/T_i$, SF-BF places all stations. Conversely (cf. Proposition 2), a station of level ≥ 1 must satisfy $d' \le t_0(1-U_0)$ where U_0 is the lower-level utilization: the threshold is tight up to the requesting station's own density share.*

Proof sketch. A failed request q at level j implies all free blocks are smaller than q; by Lemma 4 the free measure is below $2^j q$. But the free measure within the level-j window is at least $t_0 2^j(1-U) \ge 2^j q$, a contradiction. Full proofs are in the technical report, and all lemmas are additionally machine-checked on $\sim$3,000 randomized harmonic instances. □

Theorem 1. *Under the granularity assumption $d_i \le t_0/4$, Harmonic-Greedy (analysis variant, $\rho^\star = 1/4$) returns a feasible schedule of cost at most $8 \cdot \mathrm{OPT}$.*

Corollary 1 (candidate-argmin anchor)**.** *The rounded cost, as a function of the anchor phase $\theta \in [0,1)$ ($t_0 = T^{\min}2^{-\theta}$), is piecewise strictly decreasing with minima exactly at the folded candidates; hence the candidate-argmin anchor satisfies $\min_{\text{cand}} C \le \mathbb{E}_\theta[C] = (1/\ln 2)\sum_i a_i T_i$, improving Theorem 1 deterministically to $4/\ln 2 \approx 5.77$.*

Theorem 2 (unconditional)**.** *Without any granularity assumption, scaling the relaxation solution by $s = 6$ (which simultaneously enforces the density budget and, via Lemma 2, the long-SP margin $t_0 > 3d_{\max}$) yields a 12-approximation, and $6/\ln 2 \approx 8.66$ with the candidate-argmin anchor. The constant 6 is optimal for this accounting.*

Proposition 3 (peak AoI)**.** *All of the above holds verbatim for weighted peak AoI: by Proposition 1 the objective is again linear in T with coefficients $a_i^{\text{peak}} = w_i/p_i$.*

For $N = 2$ the problem admits an *exact* solution, with a structural twist:

Theorem 3 (exact N=2)**.** *Feasibility forces a rational period ratio $T_2/T_1 = p/q$ with $T_1 \ge q(d_1 + d_2)$. For each q the cost $J(r,q) = (a_1 + a_2 r)\max(T_1^{\min}, T_2^{\min}/r, q(d_1{+}d_2))$ is quasiconvex in $r = p/q$ – strictly decreasing while the $T_2^{\min}/r$ term is active, then strictly increasing – so its integer-numerator optimum is one of the two grid neighbors of the continuous minimizer, and q ranges over a finite, explicit set; this yields an exact $O(Q)$ algorithm (verified against exhaustive search on 4,000 instances). Notably,* non-integer *ratios are strictly optimal on an open set of instances – both energy floors can bind only at a fractional ratio – so the harmonic restriction has a real price already at $N = 2$, precisely the rounding loss our analysis charges for. Full proof: appendix of the technical report.*

VIII. Implementation in ns-3

We implement the full system in ns-3.[1] It comprises a `TwtPowerSaveManager` realizing individual implicit TWT as a subclass of the 802.11 power-save architecture, and the scheduler as a pluggable policy module. Three mechanism elements proved essential, each corresponding to a measured failure mode: (i) *deferral, not wake-on-traffic*: channel access requested outside an SP must not wake the radio – queued frames contend automatically at the next SP start; (ii) *SP-boundary queue gating*: without it, pending retransmissions keep the station awake until the fading flips (we measured a retry storm of 1.5×10^5 transmissions), erasing the energy savings. We gate by *freezing* rather than dropping: at SP end we mark the station's per-access-category EDCA queues blocked

[1] Code, data, and a reproduction guide are available at https://github.com/Haoyu2/age-optimal-twt – an ns-3 `contrib` module plus the `src/wifi` mechanism patch, with the raw sweep results, released to support artifact evaluation.

through ns-3's existing power-save block-reason mechanism (the same hook the standard PS manager uses), and release it at the next SP start. Blocked frames are not dequeued for transmission, so their sequence numbers, retry counts, and Block-Ack state are preserved untouched; the deferred updates simply ride the next SP. This restores the exact k-attempts-per-SP structure the model assumes and keeps the PHY in sleep between SPs; (iii) *TSF-anchored offsets*: per-station offsets must live on a common absolute time grid or the computed packing is meaningless. We further document protocol findings on TWT deployment: beacon-loss monitoring disassociates dozing stations; downlink frames (including ADDBA responses) buffered for a dozing station deadlock the uplink through asymmetric Block Ack state; and doze announcements are best-effort by nature – a PM indication cannot be acknowledged over a channel the AP cannot hear, which is precisely why TWT is schedule-based.

IX. Evaluation

Setup. 802.11ax, 2.4 GHz, 20 MHz, HeMcs3; 10–50 stations with heterogeneous weights, duty budgets and per-SP fading; one status update per wake interval per station; exact event-driven AoI integration at the AP; $300\,\mathrm{s}$ runs, 10 seeds, 95% confidence intervals.

Baselines. We compare Harmonic-Greedy against two baselines. *Equal-interval* gives all stations a single common period (the naive TWT schedule). *Energy-greedy* is a strong heuristic in the spirit of energy-efficiency-first TWT schedulers such as TASPER [27]: it gives each station its shortest budget-feasible period $\max(T_i^{\min}, d_{\max})$, uniformly stretches to feasibility if overbooked, and reuses our rounding, packing and safeguard. It is period-*diverse* but AoI-weight-*unaware*, so it isolates the value of our weighted period assignment: any margin of Harmonic-Greedy over Energy-greedy is attributable to that step alone. The two baselines let us separate two questions – does period diversity help (Energy-greedy vs. Equal-interval), and does AoI-weighting help beyond it (Harmonic-Greedy vs. Energy-greedy).

Energy–freshness frontier. Figure 3 sweeps the duty-cycle budgets ($8\times$ weight skew): the schedulers trace the AoI–energy trade-off, with Harmonic-Greedy dominating in the energy-constrained region where ideal-period spread exceeds an octave and coinciding with the baseline where the safeguard detects collapse.

Model validation. (Fig. 4, Table I) Across the loss grid $e \in \{0, 0.1, \ldots, 0.6\}$, measured AoI tracks $\delta + T(1/p - 1/2)$ with mean error $\sim 1\%$ (0.003% at $p = 1$ on every seed; worst single seed 4.4% at $p{=}0.4$, the geometric-tail variance of a $300\,\mathrm{s}$ run); renewal components $\mathbb{E}[G]$ and $\mathbb{E}[G^2]/2\mathbb{E}[G]$ are within 1% of T/p and $T(1/p - 1/2)$.

Period diversity vs. AoI-weighting. (Fig. 5, Table II) The two effects separate cleanly by regime. *When budgets are heterogeneous*, the per-station energy floors $T_i^{\min} = d_i/\rho_i$ already pin the periods, so "minimize every period then pack" is near-optimal: Energy-greedy captures essentially the whole gain over equal-interval and Harmonic-Greedy only

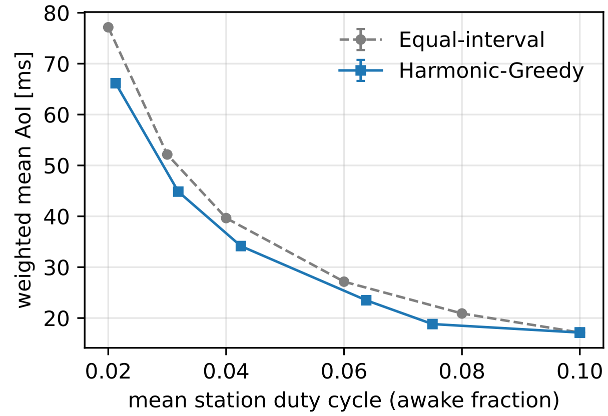

Fig. 3. Weighted mean AoI vs. mean duty cycle across energy budgets ($8\times$ weight skew, 10 stations, 5 seeds).

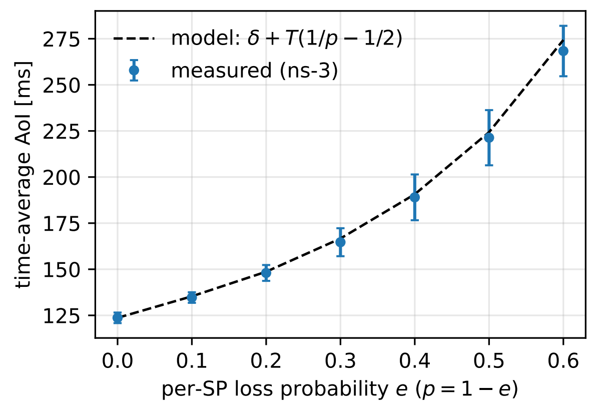

Fig. 4. Single-station validation: measured time-average AoI (3 seeds, 95% CI) vs the renewal model, across per-SP loss probabilities ($T{=}100\,\mathrm{ms}$, $d{=}8\,\mathrm{ms}$).

matches it (het-budget skew: 18.8 vs $19.1\,\mathrm{ms}$, both vs 20.9 for equal-interval; fading+skew similar). We report this plainly – period diversity, not AoI-weighting, drives those gains, and the LB column confirms both schedulers already sit near the optimum there. *When budgets are uniform*, Energy-greedy collapses to equal-interval (identical floors give identical periods), and only the AoI-weighted assignment redistributes the binding schedule density toward the stations that matter: under uniform budgets with binding density, Harmonic-Greedy wins 6.5% with weight skew (16.0 vs $17.2\,\mathrm{ms}$) and 4.3% with skew plus fading – the latter via the channel-aware coefficient $a_i \propto 1/p_i$. This is the regime that isolates the paper's contribution. Weighted *peak* AoI tracks the same pattern (Proposition 3); duty cycles equal d_i/T_i to four decimals and SPs never overlap in all runs.

Near-optimality. The LB column of Table II is the relaxation optimum – a lower bound on *any* feasible schedule. Harmonic-Greedy is the only scheduler that stays close to LB in *every* regime; Energy-greedy is near-LB when energy

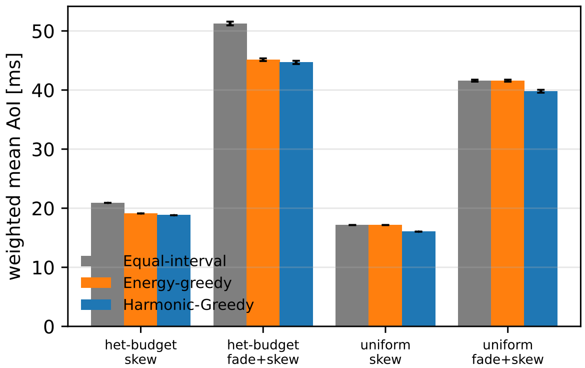

Fig. 5. Weighted mean AoI by regime (10 seeds, 95% CI). With heterogeneous budgets (left) the strong energy-greedy baseline already matches HARMONIC-GREEDY; with uniform budgets and binding density (right) it collapses to equal-interval and only AoI-weighting wins.

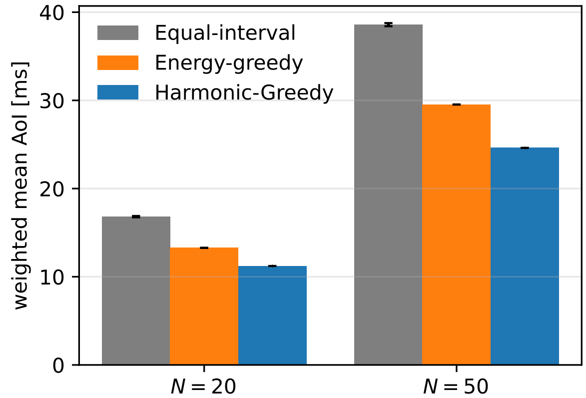

Fig. 6. Heterogeneous modulation rates (HeMcs0/HeMcs7 alternating, 700-byte updates, $8\times$ weight skew, 3 seeds): the gain decomposes into period diversity (energy-greedy over equal-interval) and AoI-weighting (HARMONIC-GREEDY over energy-greedy); 36% total at $N{=}50$.

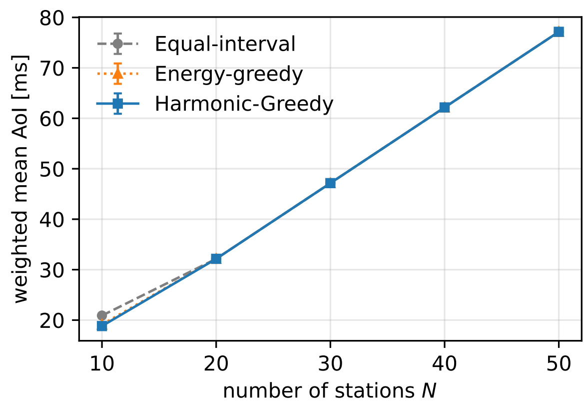

Fig. 7. Scaling with the number of stations ($8\times$ weight skew, heterogeneous budgets, 3 seeds): all three schedulers track closely once the schedule is capacity-bound (energy floors pin the periods).

floors pin the periods but drifts away once the binding density must be redistributed by weight or channel rather than by budget. The residual gap to LB is the price of harmonic rounding and contiguous (non-preemptive) packing, which Proposition 2 shows is unavoidable in general.

Scaling and decomposition. (Fig. 7, Fig. 6) The pipeline runs cleanly to $N = 50$ stations (exact duty cycles, zero SP overlap, predictions within 2%). Heterogeneous SP sizes – alternating HeMcs0/HeMcs7, 700-byte updates, airtimes differing $\sim 3\times$ – give the cleanest decomposition of the two effects at scale (3 seeds, CIs within 0.1 ms). At $N{=}50$: equal-interval 38.5 ms; Energy-greedy 29.5 ms (a 23% gain from period diversity alone, as fast stations need not wait for slow ones); and HARMONIC-GREEDY 24.6 ms (a further 17% from AoI-weighting, for 36% over equal-interval). Both ingredients contribute and HARMONIC-GREEDY beats the strong baseline outright here, because diverse airtimes create both period diversity *and* a binding density that AoI-weighting can exploit. ($N{=}20$ shows the same split: $16.8 \to 13.3 \to 11.2$ ms.)

Certified vs. practical variant. HARMONIC-GREEDY has two configurations that share the relaxation, anchor rule, and safeguard but differ in the packing stage (Sec. VI). The *certified* variant (dyadic reservations, $\rho^\star{=}1/4$) is the one Theorems 1–2 bound; the *practical* variant ($\rho^\star{=}0.9$, first-fit of true SP durations) is the one that produces the gains above. The conservative $\rho^\star{=}1/4$ is forced by the two factor-of-two losses the worst-case packing analysis must absorb in series: rounding each SP duration up to a dyadic reservation can double the schedule density, and the packing threshold (Lemma 5) admits a request only when $d' \leq t_0(1-U)$, which the count invariant satisfies for $U \leq 1/2$; starting from $\rho^\star{=}1/4$ leaves exactly the headroom for both. The practical variant instead packs the true SP durations greedily, so neither loss applies and density 0.9 is feasible – but it has no worst-case certificate. The two are thus not interchangeable: we report the practical variant's numbers throughout, and use the certified variant only to establish that the same algorithmic template admits a guarantee. The LB column quantifies the cost of this conservatism – both variants sit between LB and the baseline, the practical one near LB. Tightening the analyzed budget toward $\rho^\star{=}0.9$ is open (Sec. X).

X. CONCLUSION

TWT turns freshness-vs-energy into a schedule design problem; we gave the first formulation with AoI as the objective, the first approximation guarantees, an exact solution for $N{=}2$, and a validated open implementation. Open directions: index policies for stochastic arrivals (asymptotic optimality), exact solutions beyond $N{=}2$, AP-side schedule-aware delivery for downlink AoI, and closing the gap between the analyzed and implemented packing budgets. The latter is the most concrete: the certified $\rho^\star{=}1/4$ is conservative only because the worst-case packing analysis must absorb two compounding factor-of-two losses in series – dyadic duration rounding can double the density, and the non-preemptive packing threshold (a conse-

TABLE I
SINGLE-STATION MODEL VALIDATION (SUMMARY).

Regime	Predicted	Error
$p \approx 1$, T=100/8 ms	$\delta + T/2$	0.016%
$p \approx 1$, T=50/4 ms	$\delta + T/2$	0.065%
block fading p=0.8/0.6/0.4	$\delta + T(1/p - 1/2)$	$\leq 1.2\%$

TABLE II
WEIGHTED MEAN AOI (MS; MEAN $\pm$ 95% CI OVER 10 SEEDS), 10 STATIONS, 300 S. EQUAL = EQUAL-INTERVAL, EG = ENERGY-GREEDY (STRONG BASELINE), H-G = OURS, LB = RELAXATION LOWER BOUND (UNACHIEVABLE). TOP: HETEROGENEOUS BUDGETS – EG ALREADY NEAR-OPTIMAL, H-G MATCHES. BOTTOM: UNIFORM BUDGETS, BINDING DENSITY – EG COLLAPSES TO EQUAL, ONLY H-G (AOI-WEIGHTING) WINS.

Regime	Equal	EG	H-G	LB
het. budget, skew	20.9	19.1	18.8	15.8
het. budget, fade+skew	51.3	45.1	44.7	35.7
unif. budget, skew	17.2	17.2	**16.0**	15.2
unif. budget, fade+skew	41.6	41.6	**39.8**	35.9

quence of the same density-1 infeasibility of Proposition 2) admits requests only up to half the window. The practical $\rho^{\star}$=0.9 greedy never triggers either worst case; a tighter analysis – e.g. an amortized rather than per-request packing argument – would likely certify a density target close to it and is, we believe, the single highest-leverage next step.

REFERENCES

[1] S. Kaul, R. Yates, and M. Gruteser, "Real-time status: How often should one update?" in *Proc. IEEE INFOCOM*, 2012.

[2] R. D. Yates, Y. Sun, D. R. Brown, S. K. Kaul, E. Modiano, and S. Ulukus, "Age of information: An introduction and survey," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 5, 2021.

[3] IEEE, "IEEE standard 802.11ax-2021: Enhancements for high-efficiency WLAN," IEEE Std 802.11ax-2021, 2021.

[4] A. Kosta, N. Pappas, and V. Angelakis, "Age of information: A new concept, metric, and tool," *Foundations and Trends in Networking*, vol. 12, no. 3, 2017.

[5] Y. Sun, E. Uysal-Biyikoglu, R. D. Yates, C. E. Koksal, and N. B. Shroff, "Update or wait: How to keep your data fresh," *IEEE Transactions on Information Theory*, vol. 63, no. 11, 2017.

[6] M. Costa, M. Codreanu, and A. Ephremides, "On the age of information in status update systems with packet management," *IEEE Transactions on Information Theory*, vol. 62, no. 4, 2016.

[7] A. M. Bedewy, Y. Sun, and N. B. Shroff, "Minimizing the age of information through queues," *IEEE Transactions on Information Theory*, vol. 65, no. 8, 2019.

[8] R. D. Yates and S. K. Kaul, "The age of information: Real-time status updating by multiple sources," *IEEE Transactions on Information Theory*, vol. 65, no. 3, 2019.

[9] M. A. Abd-Elmagid, N. Pappas, and H. S. Dhillon, "On the role of age of information in the internet of things," *IEEE Communications Magazine*, vol. 57, no. 12, 2019.

[10] A. Maatouk, S. Kriouile, M. Assaad, and A. Ephremides, "The age of incorrect information: A new performance metric for status updates," *IEEE/ACM Transactions on Networking*, vol. 28, no. 5, 2020.

[11] I. Kadota, A. Sinha, E. Uysal-Biyikoglu, R. Singh, and E. Modiano, "Scheduling policies for minimizing age of information in broadcast wireless networks," *IEEE/ACM Transactions on Networking*, vol. 26, no. 6, 2018.

[12] R. Talak, S. Karaman, and E. Modiano, "Optimizing information freshness in wireless networks under general interference constraints," *IEEE/ACM Transactions on Networking*, vol. 28, no. 1, 2020.

[13] Y.-P. Hsu, E. Modiano, and L. Duan, "Scheduling algorithms for minimizing age of information in wireless broadcast networks with random arrivals," *IEEE Transactions on Mobile Computing*, vol. 19, no. 12, 2020.

[14] I. Kadota and E. Modiano, "Minimizing the age of information in wireless networks with stochastic arrivals," *IEEE Transactions on Mobile Computing*, vol. 20, no. 3, 2021.

[15] V. Tripathi and E. Modiano, "A whittle index approach to minimizing functions of age of information," *IEEE/ACM Transactions on Networking*, 2024, dOI 10.1109/TNET.2024.3452006.

[16] R. D. Yates, "Lazy is timely: Status updates by an energy harvesting source," in *Proc. IEEE Int. Symp. Information Theory (ISIT)*, 2015.

[17] M. A. Abd-Elmagid and H. S. Dhillon, "Average peak age-of-information minimization in UAV-assisted IoT networks," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 2, 2019.

[18] E. T. Ceran, D. Gündüz, and A. György, "Average age of information with hybrid ARQ under a resource constraint," *IEEE Transactions on Wireless Communications*, vol. 18, no. 3, 2019.

[19] R. Talak, S. Karaman, and E. Modiano, "Distributed scheduling algorithms for optimizing information freshness in wireless networks," in *Proc. IEEE SPAWC*, 2018.

[20] A. M. Bedewy, Y. Sun, R. Singh, and N. B. Shroff, "Optimizing information freshness using low-power status updates via sleep-wake scheduling," in *Proc. ACM MobiHoc*, 2020.

[21] E. O. Gamgam, N. Akar, and S. Ulukus, "Minimizing age of information with generate-at-will status updates and age-agnostic cyclic scheduling," *arXiv preprint arXiv:2311.18791*, 2023.

[22] B. Bellalta, "IEEE 802.11ax: High-efficiency WLANs," *IEEE Wireless Communications*, vol. 23, no. 1, 2016.

[23] E. Khorov, A. Kiryanov, A. Lyakhov, and G. Bianchi, "A tutorial on IEEE 802.11ax high efficiency WLANs," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 1, 2019.

[24] D. López-Pérez, A. Garcia-Rodriguez, L. Galati-Giordano, M. Kasslin, and K. Doppler, "IEEE 802.11be extremely high throughput: The next generation of Wi-Fi technology beyond 802.11ax," *IEEE Communications Magazine*, vol. 57, no. 9, 2019.

[25] A. Garcia-Rodriguez, D. López-Pérez, L. Galati-Giordano, and G. Geraci, "IEEE 802.11be: Wi-Fi 7 strikes back," *IEEE Communications Magazine*, vol. 59, no. 4, 2021.

[26] M. Yang, B. Li, and Z. Yan, "Target wake time in IEEE 802.11 WLANs: Survey, challenges, and opportunities," *Computer Communications*, 2025.

[27] F. Busacca, C. Puligheddu, F. Raviglione, R. Rusca, C. Casetti, C. F. Chiasserini, and S. Palazzo, "Target wake time scheduling for time-sensitive and energy-efficient Wi-Fi networks," *IEEE Transactions on Mobile Computing*, vol. 25, no. 3, pp. 3469–3487, 2026.

[28] G. Rajendran and S. Agnihotri, "Deterministic scheduling over Wi-Fi 6 using target wake time: An experimental approach," *arXiv preprint arXiv:2505.00447*, 2025.

[29] S. K. Venkateswaran *et al.*, "IEEE 802.11ax target wake time: Design and performance analysis in ns-3," in *Proc. Workshop on ns-3 (WNS3)*, 2024, doi:10.1145/3659111.3659115.

[30] E. Mozaffariahrar, M. Menth, and S. Avallone, "Implementation and evaluation of IEEE 802.11ax target wake time feature in ns-3," in *Proc. Workshop on ns-3 (WNS3)*, 2024.

[31] M. Nurchis and B. Bellalta, "Target wake time: Scheduled access in IEEE 802.11ax WLANs," *IEEE Wireless Communications*, vol. 26, no. 2, 2019.

[32] C. L. Liu and J. W. Layland, "Scheduling algorithms for multiprogramming in a hard-real-time environment," *Journal of the ACM*, vol. 20, no. 1, pp. 46–61, 1973.

[33] R. Holte, A. Mok, L. Rosier, I. Tulchinsky, and D. Varvel, "The pinwheel: A real-time scheduling problem," in *Proc. Hawaii Int. Conf. System Sciences (HICSS)*, 1989.

[34] C.-C. Han and K.-J. Lin, "Scheduling distance-constrained real-time tasks," in *Proc. IEEE Real-Time Systems Symposium (RTSS)*, 1992.

[35] M. Y. Chan and F. Chin, "General schedulers for the pinwheel problem based on double-integer reduction," *IEEE Transactions on Computers*, vol. 41, no. 6, 1992.

APPENDIX

Fix a unit $u > 0$ with $t_0 = 2^L u$. A *block of size* $s = 2^k u$ is an interval $[rs, (r{+}1)s)$, $r \in \mathbb{N}$ (aligned). Periods are

$T_i = t_0 2^{j_i}$ (level j_i, window $W_j = t_0 2^j$); reservations are $d'_i = 2^{k_i} u \le t_0$; $U = \sum_i d'_i/T_i$. SF-BF processes levels in increasing order, duplicating the free set at each level transition (valid: everything placed so far is W_j-periodic), and serves each request from a minimum-size sufficient free block, carving from the left.

Lemma 6 (staircase carving)**.** *Carving a request q from the left end of a block $B = [b, b+s)$, $q \le s$, leaves $[b+q, b+s)$, which is the disjoint union of exactly one aligned block of each size $q, 2q, \ldots, s/2$ (empty if $q = s$).*

Proof. With $s = 2^M q$, define $I_l = [b + 2^l q,\, b + 2^{l+1} q)$ for $l = 0, \ldots, M-1$. These are disjoint with union $[b+q, b+s)$, and I_l has size $2^l q$ and left endpoint $b + 2^l q$, a multiple of $2^l q$ since b is a multiple of $s \ge 2^{l+1} q$. ☐

Lemma 7 (count invariant)**.** *(i) A successful SF-BF allocation creates new free blocks only at sizes whose current count is* 0*. (ii) At the start of level-j processing, every size has count at most 2^j. Both bounds are tight.*

Proof. (i) Let q be the request and b the minimum size of a free block with $b \ge q$. The staircase sizes of Lemma 6 lie in $[q, b)$; by minimality of b no free block has size in $[q, b)$, so each created size moves from count 0 to 1. (ii) Induction: level 0 starts from the single block $[0, t_0)$; by (i) the maximum count never increases during a level; the transition duplicates the free set, doubling counts. Tightness of (ii): when intermediate levels carry no requests, doubling compounds unimpeded (verified constructively). ☐

Lemma 8 (failure bound)**.** *If a level-j request q fails, the free measure satisfies $|F| < 2^j q$.*

Proof. All free blocks have size $\le q/2$; distinct sizes are $q/2, q/4, \ldots, u$, each with count $\le 2^j$ (Lemma 7), so $|F| \le 2^j (q - u) < 2^j q$. ☐

Proof of the packing threshold (Lemma 5 in the paper). Suppose a request $q = d'_i$ at level j fails. The occupied measure within $[0, W_j)$ is W_j times the density of the stations placed so far, at most $W_j U$, so $|F| \ge W_j(1-U) = 2^j t_0 (1-U) \ge 2^j q$ by hypothesis, contradicting Lemma 8. For the converse, generalize Proposition 2: let level-0 stations occupy measure $(1-\epsilon)t_0$ of the base window with one contiguous free gap of length ϵt_0. A station of level ≥ 1 must place its contiguous SP inside one periodic copy of the gap (it cannot straddle the level-0 pattern), so $d' \le \epsilon t_0 = t_0(1-U_0)$ is necessary, where U_0 is the lower-level utilization. ☐

Proof of Theorem 1 (8-approximation). Lower bound: any feasible schedule has disjoint SPs, hence density ≤ 1 and $T_i \ge T_i^{\min}$, so $\mathrm{OPT} - C \ge J^*$ where J^* solves (2) (with the $d_{\max}$ constraint dropped; it is also valid by Lemma 2). *Density scaling:* for $\rho \le 1$, $T^*(1)/\rho$ is feasible for density target ρ (density scales by ρ, the floors still hold), so $J(T(\rho)) \le J^*/\rho$. Run the relaxation at $\rho^\star = 1/4$. *Rounding:* round each T_i up to the grid $t_0 2^j$, $t_0 \le T^{\min}$ (factor < 2 on cost; density only decreases); round each d_i up to a dyadic divisor d'_i of t_0 (factor < 2, possible since $d_i \le t_0/4 < t_0$). The resulting density satisfies $U \le 2 \cdot \frac{1}{4} = \frac{1}{2}$, and $d'_i \le t_0/2 = t_0(1-\frac{1}{2}) \le t_0(1-U)$. *Packing:* Lemma 5 places everything; each station transmits only d_i inside its reservation, so true duty cycles satisfy $d_i/T'_i \le d_i/T_i^{\min} = \rho_i$ – reservations cost no energy. *Chain:* cost $\le 2 \cdot J(T(1/4)) \le 2 \cdot 4J^* \le 8\,(\mathrm{OPT} - C)$. ☐

Proof of Corollary 1 (candidate-argmin anchor). For $\theta \in [0, 1)$ let $t_0(\theta) = T^{\min} 2^{-\theta}$ and $\ell_i(\theta) = 2^{\lceil z \rceil - z}$, $z = y_i + \theta$, $y_i = \log_2(T_i/T^{\min})$, the per-station rounding factor. (a) For θ uniform, $\mathrm{frac}(z)$ is uniform and $\mathbb{E}[2^{\lceil z \rceil - z}] = \int_0^1 2^{1-f} df = 1/\ln 2$; by linearity $\mathbb{E}_\theta[C(\theta)] = (1/\ln 2) \sum_i a_i T_i$. (b) On any interval free of breakpoints (θ with $\mathrm{frac}(y_i + \theta) = 0$ for some i), every $\mathrm{frac}(y_i + \theta)$ increases without wrapping, so every ℓ_i – and hence C – strictly decreases. As θ approaches a breakpoint from below, the wrapping station's factor tends to 1, its value *at* the breakpoint (an exact power of two rounds to itself), so C is left-continuous there and attains its infimum at a breakpoint. The breakpoints are $\theta_c = 1 - \mathrm{frac}(y_c)$, i.e. $t_0(\theta_c) = T_c/2^{\lceil y_c \rceil}$: exactly the folded candidates. (c) $\min_{\mathrm{cand}} C = \min_\theta C \le \mathbb{E}_\theta[C]$. ☐

Proof of Theorem 2 (unconditional). By Lemma 2 the relaxation may include $T_i \ge d_{\max}$. Set $\hat{T} = 6\,T^{*\prime}$: density $\le 1/6$, every period $\ge 6 d_{\max}$, cost $6 J^{*\prime}$. Any folded-candidate anchor satisfies $t_0 > \hat{T}_{\min}/2 \ge 3 d_{\max}$. Round periods (factor < 2, or $1/\ln 2$ in the candidate-argmin sense) and reservations ($d_i \le d_{\max} < t_0/3 < t_0$, factor < 2): the density becomes $U' \le 2/6 = 1/3$ and $d'_i < 2 d_{\max} < \frac{2}{3} t_0 \le t_0(1-U')$, so Lemma 5 packs. Cost $\le 2 \cdot 6\,J^{*\prime} \le 12\,\mathrm{OPT}$. Optimality of $s = 6$ for this accounting: the packing condition reads $2 d_{\max} \le \frac{s}{2} d_{\max}\,(1 - \frac{2}{s})$, i.e. $s \ge 6$. ☐

Write $J = a_1 T_1 + a_2 T_2$, $T_1 \le T_2$, energy floors $T_1^{\min}, T_2^{\min}$.

Lemma 9. *Any feasible two-station schedule has T_2/T_1 rational.*

Proof. If T_2/T_1 is irrational, $\{jT_2 - kT_1\}$ is dense in $\mathbb{R}$, so some difference of SP start times lies in $(-d_1, d_2)$: the SPs overlap. ☐

Lemma 10. *Let $T_2/T_1 = p/q$ reduced, $p \ge q \ge 1$. The schedule is feasible iff $T_1 \ge q(d_1 + d_2)$.*

Proof. Since $\gcd(p, q) = 1$, the starts of station 2's SPs modulo T_1 form q equally spaced phases (spacing T_1/q). Station 1's contiguous SP cannot straddle an SP of station 2, so it must fit in one inter-phase gap of length $T_1/q - d_2$: feasibility $\iff T_1/q - d_2 \ge d_1$. Sufficiency: place ϕ_1 in any gap. ☐

Theorem 4 (exact N=2)**.** *Let* $J(r, q) = (a_1 + a_2 r) \max(T_1^{\min},\, T_2^{\min}/r,\, q(d_1+d_2))$ *and*

$$r_q^* = \max\Big(1,\ \frac{T_2^{\min}}{\max(T_1^{\min},\, q(d_1+d_2))}\Big), \quad Q = \Big\lceil \frac{J(\lceil r_1^* \rceil, 1)}{(a_1+a_2)(d_1+d_2)} \Big\rceil.$$

Then the optimum equals

$$\min_{q\le Q}\ \min_{\substack{p\in\{\lfloor q r_q^*\rfloor,\lceil q r_q^*\rceil\}\\ p\ge q}} J(p/q,q),$$

an $O(Q)$ exact algorithm.

Proof. Lemmas 9–10 reduce the problem to $\min J(p/q,q)$ over reduced rationals with the stated feasible region (for fixed r and q, the optimal T_1 is the active maximum). For fixed q, $J(\cdot,q)$ is quasiconvex on $[1,\infty)$: while $T_2^{\min}/r$ is the active maximum, $J=a_1T_2^{\min}/r+a_2T_2^{\min}$ strictly decreases; once the maximum is constant in r, J strictly increases. Hence the integer numerator minimum is attained at one of the two grid neighbors of the continuous minimizer r_q^*. Finally $J(r,q)\ge(a_1+a_2)\,q(d_1+d_2)$, so all $q>Q$ are dominated by the $q=1$ candidate defining Q. (Non-reduced fractions replicate a smaller-q ratio with a tighter packing floor and are dominated.) Verified against exhaustive search on $4{,}000$ random instances. □

Remark 2. Non-integer ratios are strictly optimal on an open set of instances: with $T_1^{\min}=10$, $T_2^{\min}=15$, $d_1+d_2=1$, the ratio $3/2$ achieves $10a_1+15a_2$ with both energy floors tight, while the best integer ratios give $15a_1+15a_2$ (m=1) or $10a_1+20a_2$ (m=2). The harmonic restriction of HARMONIC-GREEDY thus has a real price already at $N=2$ – exactly the rounding loss the approximation analysis charges for.